

# ripwire

The ripgrep of AI context.

 *Paddle out with a map.*   
 *See the rip before you're in it.* 

A zero-runtime-dependency C++23 CLI that maps any codebase into a ranked, deterministic call graph for coding agents — and puts a tripwire on every claim it emits.

## rip — the speed half

Structural answers in tens of milliseconds, from a call graph it built itself.

## wire — the other half: honesty

Unprovable totals ship labelled as floors. A zero means none found — never none exists.

Every truncation is disclosed, and every guess says how many it chose from.

// the problem

# Your agent reads the whole library to answer one question

Blind grep, then whole-file reads. Most of what enters the context window is never used — it is paid for anyway, on every step of every task.

And bigger context is not better context: irrelevant text actively degrades the answer. The job is selection — a small, ranked, structural map beats a pile of open files.

Speed caveat travels with the number: ripwire answers from a warm, pre-built index, and round 4 measured its index cost too — 0.31 s, tabulated beside the competitors' rather than omitted. Measured 2026-08-08, before 0.6.0's performance work; not re-measured since.

**0.108 s**

median answer, warm index — round 4, all arms one machine one day, N = 60

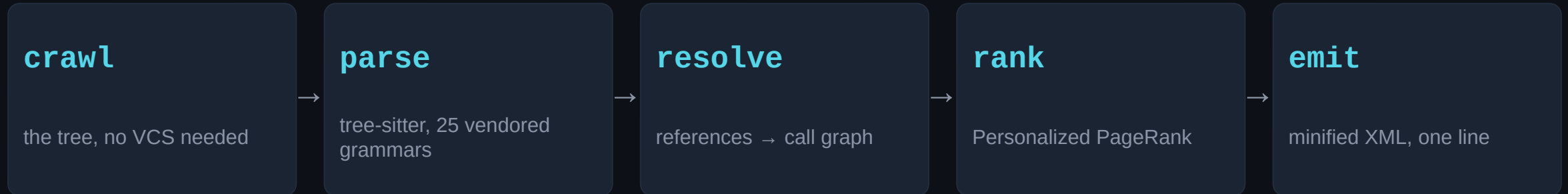
**–39.4%**

token ceiling vs the un-routed baseline — LocBench cost ledger, N = 243

// one binary, the whole pipeline

# Crawl → parse → resolve → rank → emit.

## Deterministic, end to end.



### byte-identical

two runs, same bytes — a gate on every push, not a tendency; warm equals cold

### zero runtime deps

CMake + a C++23 compiler; builds with the network off — vendored everything

### the languages

Rust · C++ · ObjC/C++ · C · Metal · CUDA · Python · Go · Swift · TypeScript · JavaScript · Java · Kotlin · Ruby · PHP · Lua · Elixir · Dart · GDScript · Bash · C# · JSON · TOML · YAML · Markdown — 25 vendored grammars, Astro frontmatter on the TypeScript one; markdown headings are real symbols

### agent-native

an MCP server and 185 long flags behind one `--help` that is always the authority

// every feature, one screen

# Seven families of questions it answers

|                 |                                                               |                                                                                                                                               |
|-----------------|---------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|
| understand cold | <i>What is this repo, and what matters in it?</i>             | --for --tree --lego --exemplar --recall --pack-task --token-budget                                                                            |
| navigate        | <i>Who calls this? Safe to change or delete? Which tests?</i> | --callers --callees --uses --impact --path --connect --affected --situ --test-gate --from-trace --pattern --safe-delete                       |
| detail ladder   | <i>Show me more — but only where it pays.</i>                 | --detail --pack-signatures --outline --expand --compress                                                                                      |
| quality & risk  | <i>Where is the risk, and did I just add some?</i>            | --quality-panel --quality-delta --dmm --biggest-first --ensemble --context-ratio --nonlocal-state --field-affinity --hotspots --lint --clones |
| self-diagnosis  | <i>Is my setup actually working?</i>                          | --doctor --skipped                                                                                                                            |
| security        | <i>Is this agent skill file safe to install?</i>              | --scan-skill --scan-skills                                                                                                                    |
| knobs & modes   | <i>Shape, format, cache, budget.</i>                          | --json --format --mcp --lsp --max-memory                                                                                                      |

--help is generated from the binary's own flag table — 185 long flags; docs/COMMANDS.md carries an entry for every one of them, 163 with a recorded invocation and its output

```
// reach for it at the moment, not the manual
```

# The reflexes: which verb fires when

## Landing cold on a task

```
--for "<task>" · --pack-task
```

ranked, budgeted context in one call

## Stack trace in hand

```
--from-trace
```

frames mapped to symbols, innermost body included

## “Is it safe to change X?”

```
--impact · --uses
```

the whole blast radius, not just 1-hop callers

## Just edited a symbol

```
--edit-check
```

did the contract change, who breaks — ~ms, warm

## About to write a helper

```
--exemplar · --grep
```

imitate the house pattern; catch the duplicate early

## Landing several branches

```
--merge-scout
```

pairwise conflict sites + a landing order

## Before you call it done

```
--quality-delta · --test-gate
```

only what you made worse; exactly which tests to run

## Handing the area on

```
--handoff · --note-add
```

the verified-vs-heuristic packet the next agent needs

and `--pr-context` when you are reviewing someone else's diff — per-file blast radius, tests-to-run, hotspot flags

// new since 2026-08-15 — each figure re-derived on the binary that ships

# What 2026-08-15 to 2026-08-23 added

## --pattern

*search by code SHAPE, not by node kinds*

foo(\$X, ...) across 13 grammar objects / 11 languages. On this repo VERIFY(\$X) finds 116 hits over 896 eligible files, each with its enclosing symbol. A pattern no served grammar resolves REFUSES (exit 1) instead of reporting hits=0.

## --safe-delete

*"can I delete this?" in one call*

callers + transitive radius + every read/write/import site + how much of that radius a test reaches. risk= names what was FOUND — never a go/no-go verdict.

## --impact importers=

*the second, weaker reach*

the files that include/require a file defining SYM, with their own cap pair — NEVER summed into reaches=, because files and symbols are different units. IngestResult: reaches=0, importers=70.

## compact --for bundles

*bodies were 52.7% of every conceptual byte*

the subtoken+body route now ships the ranked map plus one-hop edge context, disclosed as bundle=compact bodies=0. 15-query class B: 184,857 B → 95,256 B, -48.5%, all 11 decisive markers still present. --auto-bodies is a permanent opt-out.

## corroborated callers

*shared= on --pack-task rows*

a neighbour reached by four of the bundle's anchors sorts above one reached by a single anchor — omitted at 1, which is what every 1-hop row satisfies anyway.

## per-rule --lint roll-up

*a capped view stops hiding whole rules*

on this repository 16 of the 31 rules that fired contribute ZERO shown rows — 588 findings whose only evidence is their count=. A rule no corpus language registers carries applicable=0, so its zero is inertness, not a measurement.

**And two languages: PHP and Lua**, each shipped with its floor stated rather than implied — PHP's run-time dispatch (\$fn(), call\_user\_func, \_\_call) names its callee at run time and is a declared floor; a Lua corpus reports no inheritance edges, because metatable inheritance has no syntax to read.

// new since 2026-08-23 — each figure measured, dated, and re-derivable

# What 2026-08-23 to 2026-08-30 added

## --slice-flow

*cross-statement data-flow slicing  
(ARISE, arXiv:2605.03117)*

The paper's own slicer — reaching-definition edges, a seed plus a direction, a bounded BFS that stops at function boundaries — implemented and MEASURED on a registered fix-commit protocol: flow rows lift function-level added-line recall 0.163 → 0.198 at 25% of --expand's whole-body bytes (whose recall is 1.0 by construction). 7 commits / 38 instances, cpp only — a thin corpus, reported as thin.

## --slice, first numbers

*the 2026-08-28 registered contract,  
finally measured*

Per-variable line-recall 0.726, hit-all rate 0.632 — a FLOOR under a noisy relevance oracle: every inspected miss was the protocol's word-regex matching identifiers inside comments and string literals, occurrences the slicer correctly refuses to call variable uses. No inspected instance showed a real occurrence the slice dropped.

## ~1350× on --expand

*the secret-redaction sweep was  
quadratic in LINE length*

One selector in a 2.1 MB / 768-line minified yarn bundle NEVER completed — killed at 1,343.9 s of user CPU with an empty output file; 196.7 s is the only clean lower bound — and now answers in 0.46 s warm. The 20 KB single-line fixture: 23.02 s → 0.017 s. A pure memoization, so an output no-op: 120/120 invocations byte-identical; gated on behaviour, the timings a ledger row and never a red-CI threshold.

sources: docs/EVALS.md §--slice-flow (registered protocol + per-instance ledger) · bench/PROFILE.md 2026-08-30 — no number travels without its caveat

// new since 2026-08-30 — the window this deck did not cover until now

# What 2026-08-30 to 2026-09-06 added

## **.gitignore, honoured by default**

*named for ripgrep, whose defining default this is*

The crawl walked ignored files; in a git work tree it now consults git's own rules and skips what the repository already declared uninteresting. This root, three checkouts under an ignored bench/external/: files= 8,674 → 1,522, cold 2.81 s → 0.52 s, warm 0.63 s → 0.10 s — and --no-ignore restores the previous walk exactly. Nothing drops silently: ignored\_files= is exact, ignored\_dirs= says the walk STOPPED there so those contents are UNKNOWN rather than zero, and both are ABSENT when the rules dropped nothing — so a tree with nothing ignored is byte-identical to what it produced before.

## **--html --color-by**

*the map, rendered —  
no server, no CDN*

One self-contained HTML file: a force-directed call graph you click to recentre. --color-by=lang|community|cx|churn|tested sets the INITIAL colour only — the page embeds all five and keeps a live selector, so switching lens costs no second run. One five-stop blue-to-orange scale rather than the usual green-to-red, so reading it never depends on telling red from green.

## **--eval-retrieval, re-sampled**

*the sampler was measuring  
the corpus, not the ranker*

For fourteen months the gold set was the first 150 doc-commented symbols in PATH order — and bench/ sorts before src/, so adding a docstring to a benchmark harness moved a published number without touching ranking code. One 60-symbol probe, one corpus, only its PATH varied: 0.105 and 0.356 MRR from spelling alone. Now exhaustive — population=3011 scored=3011 rule=exhaustive — and EVERY retrieval figure this project publishes was re-derived under it.

## **--plan-lanes**

*which lanes would collide,  
BEFORE a line is written*

Where --merge-scout says "these branches already conflict", this says "these lanes WOULD conflict if assigned this way" — no ref to resolve, no re-ingest. Each lane now also carries an advisory execution profile (model + reasoning effort) that labels its own evidence basis="structural-only" and versions its policy separately from the schema, so a recommendation cannot quietly claim more than the structure it read.

CHANGELOG.md [Unreleased] carries the ignore-default measurement and its ledger in bench/PROFILE.md · the sampler negative is docs/EVALS.md §7, published as a counterexample against our own published numbers

// 0.6.0 – everything since v0.5.0 (tagged 2026-09-07); each figure names its PR

# 0.6.0: faster where it hurt, clearer where it stops

## two new languages

### Dart

the 23rd grammar, by @calvinchengx: 71,726 symbols from flutter/packages' 3,706 .dart files, none among its 289 degraded parses (#75, #106)

### Kotlin

by @xCatG: calls resolve between Kotlin and Java, both ways; no sanitizer finding across 501 real .kt files, deterministic on ~9,500 more (#126)

### any language

Dart's landing caught six per-language arrays sized off the last enum by hand: an appended language dropped out of --skipped's census silently. Fixed for all.

## Rip'n Fast

**159.7 → 9.2 s**

warm --grep on llvm-project: an inheritance cone rebuilt on every still-ambiguous receiver-typed call is now built once per type (#83)

**-19.9% CPU**

cold parse on llvm-project, 194.1 → 155.6 s: child walks that went quadratic now use a cursor, output byte-identical (#127, #130)

**-27.6% CPU**

warm --pack-task on go, 8.13 → 5.88 s: the tokenizer rebuilt on one header of NEON, AVX2 and scalar string kernels (#127)

## honest where it counts

**2,107 → 3**

false callers of memgraph's SafeString::move: a std::move call no longer binds to your own code (#134)

**12/12 → 8/12**

commits --quality-delta gated, of 12 landed: true-positive share 2% → 12%, with 0 wrong rows (#127)

**13/25 → 0**

harmful --help-task recommendations on the adversarial prose set; precision 0.797 → 1.000 (#127)

## fewer tokens, nothing hidden

**46,385 → 4,473**

tokens for --help: one line per flag now, and every disclosure is still one call away (#92)

**6 → 50**

--handoff symbols per code file (12 per prose file): containment 16% → 54%, purely additive (#127)

**pages, not cuts**

--doc-drift, --flags, --flip and --situ disclose their cuts and page; --situ lists every tests-to-run row (#127)

upgrade note: prebuilt x86-64 binaries now need an x86-64-v3 (AVX2-class) CPU, RHEL 10's own floor – the installer checks before it downloads and names what is missing; arm64 uses NEON (#127, #137, #138)

// 0.6.1 — everything since v0.6.0 (tagged 2026-09-07); each figure names its PR

# 0.6.1: it resolves what it names, and says what it cannot prove

## languages that resolve, not just parse

**module · name ·  
arity**

Elixir calls resolve to the module, name and arity they name, not by name alone: lexical aliases, filtered imports, default arguments, pipes, captures, delegates. Macro expansion and unquote stay documented limits (@henry-hz, #207)

**none → 79%**

of occurrences matched on spring-petclinic: SCIP encodes a range two ways and ripwire read only the deprecated one, so every scip-java index was silently ignored and --scip changed nothing (@dpunosevac, #198)

## it says what it could not prove

**89 of 4,322**

answers shrank over every file:name selector on ripwire's own tree that selects only declarations — and NONE grew. A header selector kept a definition only when its file is the header or includes it, resolved path-precisely (#173)

**nine listing  
verbs**

now count what the proof dropped as unproven\_defs=; all but two had answered from the declaration alone and printed a clean zero (#190, #195)

**not a safe edit**

an incompatible="0" beside unproven\_defs= is now STATED to be an incomplete read, on every verb that resolves a focus symbol, --edit-check included (#210)

## every number carries its definition

**graph\_unindexed=**

shipped in 0.6.0 with no definition on three verbs, and under a compact legend on every XML verb but one (#169)

**and eight more**

declined\_calls=, unproven\_defs=, pr\_iters=, the map header's counts, --impact's blast-radius counts, --safe-delete's verdict fields and the community structure counts: each defined now (#185, #189, #203)  
the compact byte pins now follow the definitions, instead of a definition being trimmed to fit one pin; a budgeted --for that drops legend clauses NAMES the attributes it dropped (#174)

**pins follow prose**

## it fits in memory at llvm scale

**114 MB → 368 KB**

the declined-call index on llvm-project. Candidate lists were stored once per call, so the structure grew with calls x candidates: 27.9 M entries. One copy per DISTINCT list makes it 9,879 lists and 62,359 entries (#208)

**every byte the  
same**

counts and output are unchanged by that change — the win is the structure, not the answer (#208)

the release ran against instruments registered before the work began: a frozen bank of 30 retrieval questions answered in ONE call on a 2,066-file corpus pinned at one commit, a six-step follow-up ladder with no model in the loop, and a separately registered held-out draw so a round cannot be tuned onto the bank it is graded on

// since v0.6.1 (2026-09-14): 0.6.2-0.6.5 and main; each row names its PR

# Since 0.6.1: more places it runs, more code it answers for

## where and how it runs

**Windows x64**

builds under clang-cl and MSVC cl.exe; a release zip ships as a preview. The gate suite does not run there yet (@lennix1337, #44)

**--lsp**

a read-only navigation LSP server over stdio: definition, references, document and workspace symbols, hover (@mpapis)

**65% · exit 5**

a zero-config memory guard: past 65% of RAM or the cgroup limit, a disclosed partial map or exit 5, not a swap storm; --max-memory moves the line (#350)

## better answers

**63 → 2,083**

functions --quality-panel measures on a 583-file TypeScript repo: a name bound to an arrow or function literal now owns its body (JS, TS, Lua, Python)

**one row per def**

--metrics: overloads and #ifdef twins each get their own row and metrics, instead of sharing one body's

**Ruby, from the community**

class Child < Parent is an edge, so --lego answers for Ruby (#325); RSpec's described\_class pins to the class under test (#338), @andriyturnikov

**72.8% of vue-core**

call sites had no caller, measured before the fix: a call outside every named function now has a module-scope owner (#60)

## honest answers

**rank, then cut**

capped verbs keep their strongest rows and name what they dropped (0.6.3); three answers that cut silently now say what they left out

**shallow="1"**

on a depth-1 clone, the CI checkout default, the history verbs say the history is shallow instead of calling the boundary commit a root

**five false answers**

--flags, --doc-drift, --from-trace, --connect and --dead-code each answered falsely on a real repository; each is fixed

## quality you can act on

**placeholder**

the 11th --quality-delta kind: the stubs and TODOs a change adds, per symbol. It lists them and never gates

**d= hop depth**

--impact lists the blast radius nearest first, so a cut page drops the deepest rows first; by\_depth= says where it stopped

**--mcp-tools**

the MCP server lists only the tools you name, or a core profile of 8; the full 33 stay the default, byte for byte

upgrade note: parser 143 · cache 28 — a cached index from an earlier build re-indexes once on first use

// Rip'n Fast, at scale – llvm-project, 182,555 files, as the instrument

# Three bugs found at llvm-project scale

## **0(C<sup>2</sup>) child walks**

An indexed `ts_node_child( n, i )` loop re-walks the child list on every call, so wide, flat lists (comment runs, C/C++ include guards) go quadratic. #127 moved 24 walks to one cursor helper (18 arms red first at 11–125×); #130 moved 22 more (13–126×).

**194.1 → 155.6 s**

cold parse CPU, n = 1  
–19.9% · wall –25%

## **the inheritance cone**

Rebuilt on every still-ambiguous receiver-typed call: 86,667 cones for 2,984 distinct types, 143 s of a 154 s run. Each type's cone is now built once, and default maps are byte-identical on go and llvm-project.

**159.7 → 9.2 s**

warm --grep  
default map 248 s → 10 s

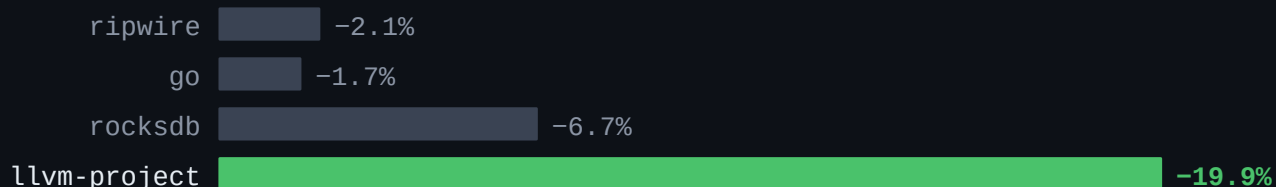
## **the cache that evicted itself**

One llvm root needs 1.76 GB of cache, and the 2 GB oldest-first sweep was evicting that root's own index. The sweep now pins the root you are working in, and two key builders that hashed one root with different seeds share one key.

**274 → 26 s**

a repeated --for, CPU

the same round's cold parse, CPU Δ by corpus (median of 5; llvm-project n = 1)



## **Why llvm-project**

*A scale test that is also the real thing. Built on or with LLVM:*

### **Clang (C, C++, Objective-C)**

Chromium's only supported compiler; the Android NDK's; Apple Clang in Xcode; the PS4 toolchain

### **Swift · Rust · Julia · Zig · Kotlin/Native**

each has an LLVM backend for its code generation

### **CUDA**

NVIDIA's NVVM compiler is based on LLVM, and Clang compiles CUDA too

### **Metal**

Apple's shading language: "Metal uses clang and LLVM"

### **MLIR**

part of LLVM; used by TensorFlow, OpenAI Triton and Mojo

### **WebAssembly**

Emscripten, a compiler toolchain to WebAssembly using LLVM

### **ROCm · oneAPI**

AMD ROCm's compilers fork llvm-project; Intel's oneAPI DPC++/C++ uses LLVM

LLVM received the 2012 ACM Software System Award · each item's source is in the notes

all three measured on llvm-project; no standard corpus could see the first · sources: #127, #130, #83 – quotes in the notes

// wire, in practice – three stories from 0.6.0, all merged

# Confidently wrong, caught, and fixed in the open

#134 · the resolver

## std::move had 2,107 callers

On memgraph, calls written std::move bound to the repository's lone in-repo move, SafeString::move, at full confidence and with no amb= or prov= marker. It became map row #1. In ripwire's own graph, std::min, std::max and std::sort bound fastmath and svector members.

**2,107 → 3**

callers of SafeString::move after the fix: two of its unit tests and one std::ranges::move, a disclosed gap

what catches it now

A std::-qualified call keeps only candidates inside namespace std; anything else counts toward external= and gets no edge. On ripwire's own tree, --callers=min went from 131 to 5.

test/stdqualcheck.sh · 35 checks, 19 red before

#127 · the string kernels' gate

## The probe that lied

strkerncheck asked Rosetta 2 whether it could run the x86-64-v3 kernels, using a one-add AVX2 probe. Clang folded that probe to a scalar addb, with zero ymm or VEX opcodes, so it answered ok on every runtime while the v3 slice SIGILLED on the macos-14 runner. The first fix's commit gave the wrong reason; the next one corrected it.

**0 ymm opcodes**

in the probe binary that had answered "ok" (otool)

what catches it now

The probe is now built with the floor's own -march and touches every v3 extension. The gate disassembles it and requires ymm, pdep, pext, lzcnt, tzcnt, vfmadd and vcvtph2ps; a folded probe routes to the scalar slice instead of passing.

test/strkerncheck.sh · b4ebf0bb, 150fb6d3

#127 · the audit's own tools

## The audit's instrument was wrong twice

A 14× super-linear reading turned out to be the cache evicting its own working root. And the recorded cap sweep had measured retrieval caps on a population of zero: it discarded exit codes, counted rows that emit nothing, and read back its own 10.4 MB cache blob, which "responded" to 103 of 108 caps.

**59/195 → 64/151**

cap-sensitive rows: the published split, then the split re-derived over the rows that answer

what catches it now

Every sweep row now carries a state (ok, rc=N, unparseable, unexpanded, timeout), a run where nothing answered refuses to report, and nothing it writes may land in the corpus. The cache sweep pins the root you are working in.

test/capsweepcheck.sh · 209f97a4 · 5723b2c0

every figure is quoted from its PR body or commit in the speaker notes – the tripwire covers our own instruments too

// wire, in practice – three more: the parser, the resolver, and a new language

# Three more: a semicolon, a silence, a new language

#135 · the parser

## One missing semicolon

A member macro with no ';' as a class's last line, like `EXC_NAME(Foo)` before `};`, derails tree-sitter's C, C++ and ObjC error recovery. On memgraph one file had 472 of its 487 definitions misfiled, and a 14-line function ranked #4 in --hotspots, with nothing on the row saying so.

**ccx 920 → 2**

PrintFuncSignature, 14 lines: cx=749 ccx=920 before, cx=3 ccx=2 after the re-parse

what catches it now

Misfiled definitions carry `extent_suspect=`, and --hotspots leaves them out and says so. An error-gated re-parse blanks ALL-CAPS member macros and keeps the result only if it has fewer error bytes. Lowercase and namespace-scope runs still derail: flagged, not fixed.

extentcheck.sh (55) · macroreparsecheck.sh (103)

#136 · the resolver

## 22% of calls, declined in silence

--callers answered count=0, as if nothing called. When a call names two or more same-named definitions, none in the caller's file or directory, the resolver declines to guess: the right rule, but the decline was silent. On memgraph, 22% of all call references were declined this way.

**65,516 / 295,086**

memgraph call references declined (22.2%), now shown as declined= and declined\_calls=

what catches it now

Every call reference ends in exactly one named bucket, and --pin-census prints the balance. It proved itself in the merge: #134's std:: refusals had gone uncounted, 1,495 on ripwire and 4,966 on memgraph. Now they count, and unaccounted=0.

test/declinecheck.sh · 17 languages, 104 checks

#126 · Kotlin, by @xCatG

## A new language must not change old answers

Kotlin brings calls that cross into Java. In #126's review, once retrofit's same-named Kotlin body() methods were indexed, calls to the Java Response.body were declined, and its callers fell from 279 to 5: a Java-only answer, changed by files in another language.

**279 → 5 → 279**

callers of retrofit's Response.java:body: before Kotlin, with the first Kotlin binary, after the fix

what catches it now

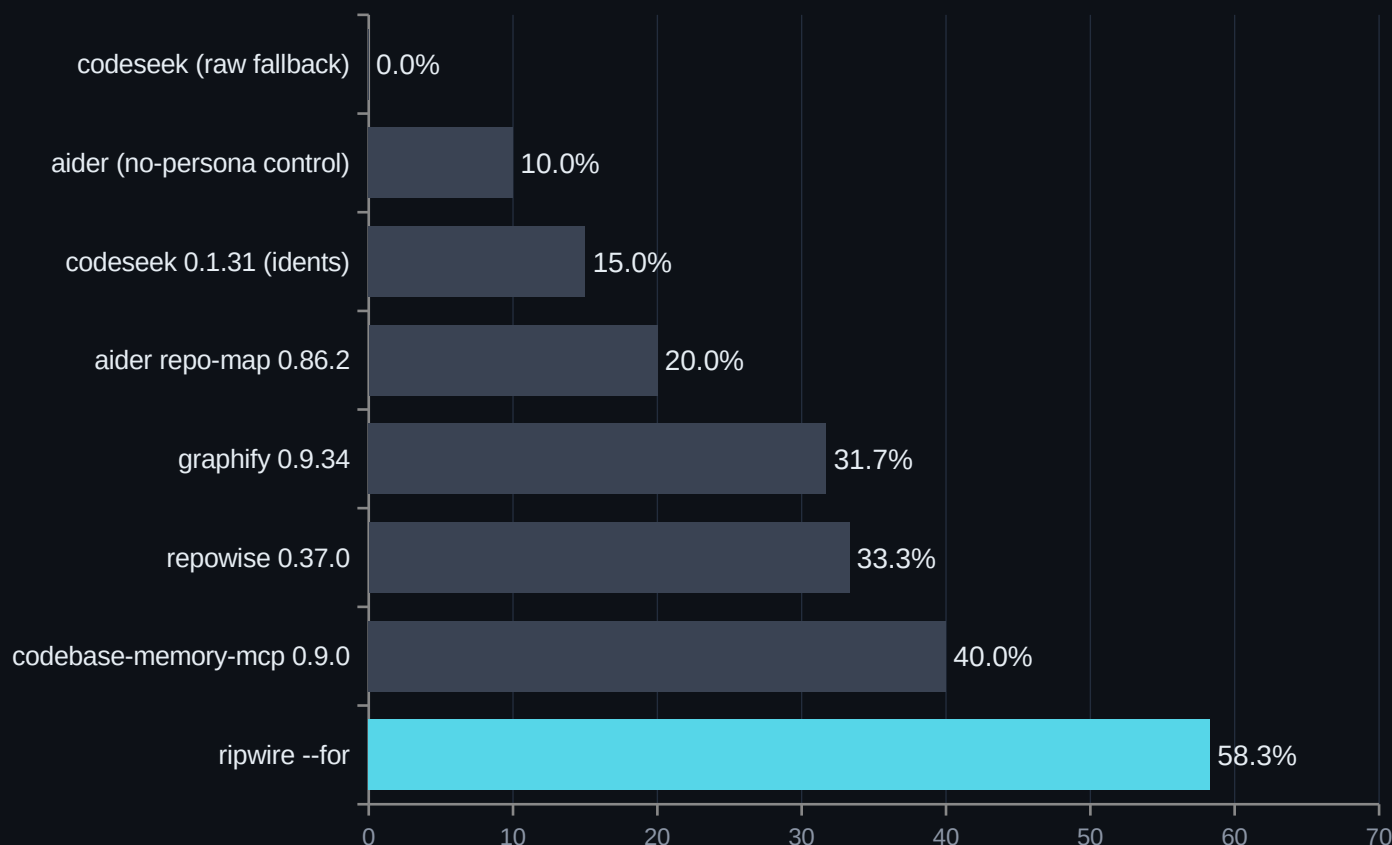
A call reaches the other JVM language only when its own language has no candidate, so adding .kt files never changes a Java-only edge. The bridge keeps its real job: a call into a name only the other language defines.

test/kotlincheck.sh §14 · the Java-only invariant

every figure is quoted from its PR body or commit in the speaker notes

// one harness, one machine, one day — N = 60 paired, zero exclusions

# Head-to-head: every competitor, one table



1.46x

the margin over the best competitor — 58.3% against 40.0%  
strict file@10

## And the index it was measured with

ripwire 0.31 s · codebase-memory-mcp 1.24 s · codeseek  
3.37 s · graphify 7.82 s · repowise 34.0 s, whose worst  
single index was **352 s**. Cold, nothing to an answer: 0.213 s.

**What re-running cost us, published:** aider moved 18.3%  
→ 20.0% on its own tie-break nondeterminism, and we  
printed the higher number. Paired, ripwire loses 2 instances  
to codebase-memory-mcp, 2 to repowise, and 1 each to  
graphify, aider and the aider control.

// scored against a third-party oracle, never against each other

# Graded by a compiler: zero silent misses

Both tools were scored against a scip-clang compiler-grade index of this repository — 68 answers, 34 blind-authored queries × --uses and --callers.

0

true silent misses — pre-fix and post-fix alike. Every imperfect answer either carried a discriminating self-flag (amb=, defs>1, external="1") or was an oracle-scope artifact where ripwire was right and the compiler could not see the file.

## What a compiler-grade index costs

|             | ripwire | Serena 1.6.2.dev0 (clangd 19.1.2) |      |
|-------------|---------|-----------------------------------|------|
| warm query  | 49.8 ms | 3,760 ms                          | ~70× |
| cold start  | 208 ms  | 8.7 s                             | ~40× |
| index build | none    | ~8 min                            | —    |
| errors      | 0 / 101 | 7 / 61                            | —    |

**Read honestly in both directions.** The LSP is more precise — site-level 0.929/0.941 against ripwire's 0.914/0.941 after the fix round, a gap of roughly 1.5 points with recall now equal. ripwire also pays ~7× the bytes per call, because its output carries self-describing legends. What it cannot do is answer a macro query at all — clangd's documentSymbol has no macro entries, which is 3 of its 7 errors — or start in less than eight minutes.

**The number that moved the wrong way, published with the rest:** over-hedging rose 18.6% → 22.6% across the fix round. Closing loss buckets added self-flags faster than it removed imperfect answers. That direction is deliberate — calibrated to warn too often rather than too rarely — but it is a cost, and it is on the slide.

// 48 paired questions, zero exclusions, both arms exit 0 on all 48

# Against a graph-database context server: 27–7–14

| class                  | n  | ripwire | GitNexus | tie | bytes ÷ it |
|------------------------|----|---------|----------|-----|------------|
| symbol lookup          | 12 | 6       | 2        | 4   | 1.35×      |
| conceptual search      | 15 | 8       | 1        | 6   | 1.23×      |
| callers / blast radius | 12 | 7       | 2        | 3   | 0.39×      |
| task orientation       | 9  | 6       | 2        | 1   | 0.06×      |
| all 48                 | 48 | 27      | 7        | 14  | 0.159×     |

## The cost of the whole sweep

**309,348 B** against 1,945,231 B. Warm median 197 ms against 1,082 ms. Index 0.25–0.45 s / 6.6–16.5 MB against 23–52 s / 391–623 MB — written into the repository, not \$TMPDIR.

**What it does better:** compact one-hop context at 0.7–1.0 KB, a depth-LABELLED blast radius, per-stage timing on every query, and two-command onboarding into eight agents.

## Failure modes, both directions

|                            |   |      |
|----------------------------|---|------|
| needed a 2nd call          | 0 | 8/48 |
| empty radius, real callers | 0 | 1    |
| malformed through a pipe   | 0 | 7/48 |

*The published numbers use the file-redirected form — the configuration favourable to it.*

**Its seven wins are named one by one, and its first-pass numbers do not exist.** Four are cost losses on an answer ripwire gets right — a by-name lookup it serves in ~1 KB where ripwire spends 3× to also hand back the body. Three are ranking misses on webpack, and two of those share one unfixed mechanism: a symbol whose NAME matches the query beats the implementation that carries it in its body. Under the improve-first rule the round's FIRST pass was never published — its product was a loss list, and what is on this slide is the state after that list was worked. One of those fixes met its pre-registered band and was reverted anyway for failing a separate standing requirement; **the loss it would have fixed is still counted against us above.**

// the rule: numbers are held back until the loss list is worked

# Every head-to-head round, and the ones that went against us

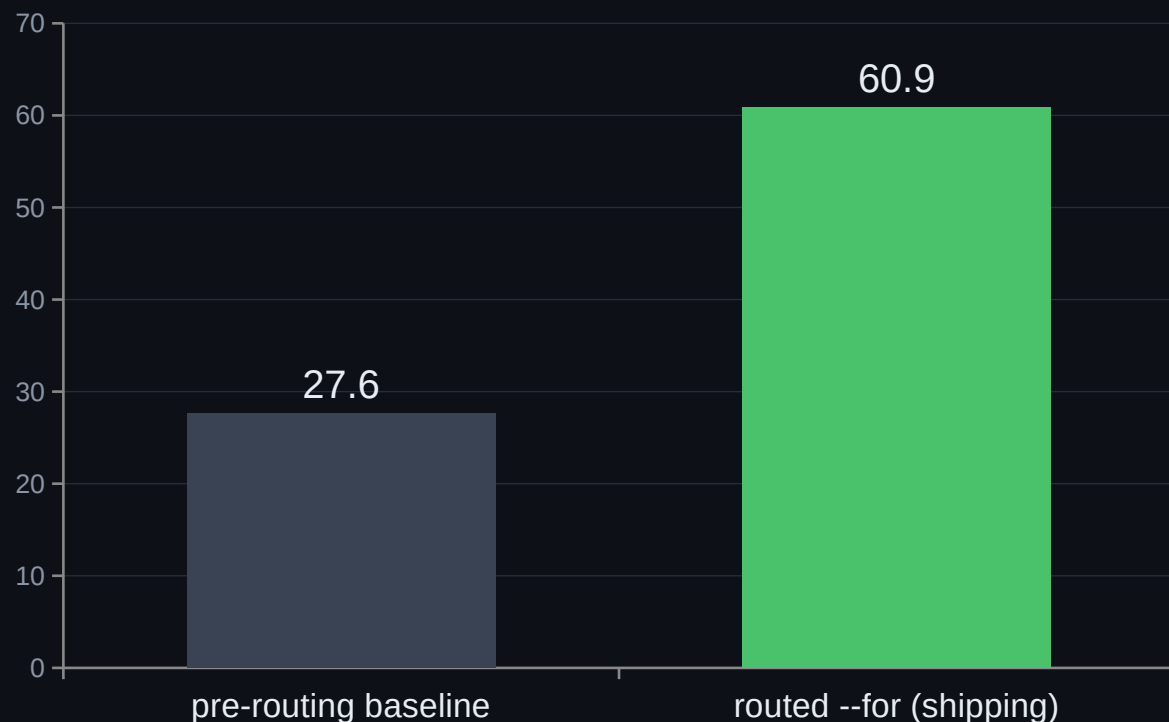
|     |               |                                                           |                                              |
|-----|---------------|-----------------------------------------------------------|----------------------------------------------|
| r1  | 2026-07-13/14 | Aider repo-map · codebase-memory-mcp · graphify           | superseded by r4                             |
| r2  | 2026-08-03    | repowise · codeseek                                       | superseded by r4                             |
| r3  | 2026-08-03    | headroom — a compression layer, so a different instrument | passed code through byte-identical           |
| r4  | 2026-08-06/08 | all five competitors, one harness, one day                | the table on the previous slide              |
| r6  | 2026-08-04/05 | agent-in-the-loop: the Codex CLI pilot                    | localization parity; +80% tokens, classified |
| r7  | 2026-08-08    | CodeGraph, loss-first — we looked for our losses          | fix REJECTED by its own preregistered band   |
| r8  | 2026-08-08    | aider again, at equal token budget                        | loss questions traced to one root cause      |
| r9  | 2026-08-09    | Serena / scip-clang oracle                                | 0 silent misses; the fix list it produced    |
| r10 | 2026-08-22    | GitNexus 1.6.9 — a graph-database context server          | 27-7-14 — its own slide, two back            |

**r7 in full, because it is the point:** the fix was built, gated green, measured — predicted 12/22 against a pre-registered band of 10–14, measured 5/22 — and reverted rather than tuned. Published as a negative in docs/EVALS.md §7.

bench/headtohead/ (r1-r4, r9) · bench/r7/ · bench/r8/ — harnesses committed even for rounds that rejected their own fix · r5 left no head-to-head record in this tree, so it is not listed · r10's method and pins are published in docs/EVALS.md §2, its pre-fix numbers nowhere

// held-out, repo-disjoint, frozen dataset

# Routing the ranker: +33 points, measured properly



**+33.33pp**

paired delta, 243 held-out instances in 78 repository clusters

**+25.00pp**

clustered-bootstrap 95% lower bound — the honest floor

**-39.4%**

token ceiling for that gain

**+3.4%**

warm latency for that gain

// compiled, not interpreted — and measured on public corpora

# Fast enough to call on reflex

**0.213 s**

cold — nothing to an answer: parse, rank,  
reply, no cache (r4)

**0.108 s**

warm query, median — against aider's 2.920  
s (r4)

**0.31 s**

index build — against repowise's 34.0 s,  
worst case 352 s (r4)

**49.8 ms**

warm query on this repo — against a clangd  
LSP's 3,760 ms (r9)

## Where the second goes

Cold profiles are tree-sitter parse plus tag-query capture — not graph math. Resolve and PageRank together are a small fraction of the run, and the ranking was never the bottleneck.

Which is why the wins came from removing work, not micro-optimising it: demand-driven side captures, five whole-AST passes fused into one pre-order walk, a calibrated per-worker parse reserve.

## Two opt-in faster builds

LTO is on by default. PGO is a script away and buys 14–25% cold.

Both are opt-in performance, not opt-in correctness: **the output stays byte-identical across build flavours** — the determinism gate runs on both, and CI builds a plain flavour too, because NDEBUD once compiled a whole class of degrade-path checks out of existence.

every card above comes from a committed harness — bench/headtohead/r4-2026-08-06/ and r9-2026-08-09/ — not from a private corpus

// how it got there — the pre-release performance sprint

# Work removed before work optimized

## file ingest

FILE\* whole-file reads; CSV counting became a byte scan — iostream out of the hot path

## capture gating

value-use capture runs only for the verbs that actually need it, not on every run

## AST pass fusion

five whole-AST side-capture passes share ONE pre-order walk instead of five

## parallel parse

query compile overlaps parse scheduling; files schedule by size — ids stay deterministic

## radix + S+tree

numeric keys for edges and scores; a bounded-scratch hot map whose fixed cap reports its own overflow

## no std::map

unordered\_dense, reserve() before ingest — no std::unordered\_map on any hot path

**Why this slide carries no numbers.** The sprint's own headline pair was measured in June 2026 on a large private C++ corpus that is not publicly reproducible, and the deck does not quote figures a reader cannot re-derive — the previous slide's numbers all come from committed harnesses. What survives the omission is the **shape**: the graph math was never the bottleneck, and every win came from doing less work rather than doing the same work faster. Same map contract throughout — byte-identical output before and after each of these changes.

the mechanisms are in bench/PROFILE.md and the commit history; the private-corpus figures they produced are deliberately not quoted

// docs/EVALS.md §5, pinned on this repository 2026-08-08 (the --for row re-pinned 2026-08-22)

# Ten everyday moments, and what each one costs

| Ask it                                 | Command                   | ripwire    | naive read | savings      |
|----------------------------------------|---------------------------|------------|------------|--------------|
| Orient me in this repo                 | ripwire .                 | ~5.6K      | ~20K–25K   | 3.6×–4.5×    |
| Where is X handled?                    | --for="..."               | ~2.1K      | ~4.9K–20K  | 2.3×–9.3×    |
| What do I already know?                | --recall="..."            | ~15K       | ~445K      | 29.2×        |
| Set me up for this task                | --pack-task="..."         | ~2.1K      | ~16K–80K   | 7.7×–37.7×   |
| Show me this one function              | --expand=SYM --top-k=0    | ~260–16.5K | ~43K–174K  | 2.6×–670×    |
| Who calls this function?               | --callers=SYM             | ~580       | ~40K–52K   | 69.2×–89.1×  |
| Is it safe to change this?             | --impact=SYM + --uses=SYM | ~1.3K      | ~18K       | 14.4×        |
| I have a stack trace                   | --from-trace=FILE         | ~1.4K      | ~124K–298K | 86.9×–208.6× |
| I changed these files — tests? radius? | --situ                    | ~410       | ~3K–132K   | 7.3×–324.2×  |
| Review this PR/diff                    | --pr-context=REF          | ~1.9K      | ~4.8K–51K  | 2.6×–27.5×   |

Figures are ~tokens (≈ bytes/4). Every row is scored same-correct-answer-or-it-doesn't-count — both sides were checked, not assumed — and every ratio is a **range**: the cheap end and the honest end of what an agent would actually read, never the single most flattering number. Absolute counts belong to the corpus AT THE PIN — .gitignore became a crawl default on 2026-09-03, which moved what a re-run reads; the commands, not the constants, are what reproduces.

// the cost lever

# Smaller answers that are also better answers

## 74.7%

fewer element bytes at top-50 — the --pack-signatures ladder,  
root-neutralised, re-derived on this repo every run

Ask for a symbol by NAME and the router uses the  
name-exact ranker:

**MRR 0.745 → 0.960**

**recall@1 61.1% → 91.3%**

name-shaped queries on src/, all 3,011  
doc-commented symbols scored, re-pinned  
2026-09-05

This figure survived its own audit: three corrections deep, re-derived  
root-neutrally after the original was shown to depend on how the  
corpus path was spelled — three spellings of one root read 18.6  
points apart before that subtraction. top-50 is the quotable number  
because the payload is top-50 whatever --top-k says.

And the pollution metric — fixture and generated paths  
contaminating results — is driven to 0.0% on the ranking lane  
while recall went UP, not down.

docs/EVALS.md §4-§5 — showcapturecheck re-derives the byte-reduction triple; `ripwire src --eval-retrieval` re-derives the ranking  
bars, scoring every doc-commented symbol rather than a sample

```
// the half no other tool ships
```

# The tripwire: honesty as an output contract

## **counts\_floor="1"**

call edges come from source text — dynamic dispatch adds no edge, so every count that cannot claim completeness is labelled a floor, in the output itself

## **a zero is a measurement**

zero means none found, never none exists — and absent  $\neq$  0

## **truncation is disclosed**

every cap, page seam and budget cut is named in the header before you read a row

## **refusals teach**

a refused query names the flag, the problem, and a working example — never a bare error

```
$ ripwire . --callers=rankGraphTeleport  
  
<callers of="rankGraphTeleport"  
  defs="1" count="7"  
  counts_floor="1">  
<s t="fn" n="getIndex" .../>  
<s t="fn" n="rankGraph" .../>  
...  
</callers>
```

**The map grades itself before it answers.** This repository's own src/: files=195 symbols=8318 edges=25147 ambiguous=9391 unresolved=2008 declined=7742.

// the honesty marks stop being a promise and become a measurement

# Four levers accepted, one rejected on purpose

## macro edges

kParserVer 48

function-like #define invocations land role="macro" edges on disclosed-degraded symbols

## fn-pointer bindings

kParserVer 49

calls through unambiguous local function-pointer bindings now resolve

## using-declarations

kParserVer 51

re-exports emit role="import" references — r9 loss bucket 1

## shadow suppression

kParserVer 52

a local variable shadowing a function name stops stealing its use-sites — r9 loss bucket 2

## RTA-lite

reverted

instantiation-filtered CHA cone: refuted TWICE — the instantiated set crossed namespaces, and narrowed calls LOST their amb= mark

The two r9 levers, against the compiler oracle: **site precision 0.9046 → 0.9136, recall 0.9285 → 0.9412.**

**The fn-pointer lever raised unresolved= by 1,039. Disclosure, not regression:** call sites the resolver now RECOGNIZES but refuses to resolve are counted instead of silently ignored.

// how it stays true

# Proven, not promised

## 664 gate scripts

the suite runs on every push — plus determinism, cache-transparency and golden contracts; the gate count itself is gated against the runner's own loop

## byte-identical, always

two runs over the same tree produce the same bytes; warm equals cold. Enforced in CI, twice — Release AND a plain flavour, because NDEBUB once blinded a whole class of checks

## differential refactoring

a refactor must prove it changed nothing observable: two binaries, hundreds of argv vectors, stdout + stderr + exit codes byte-identical

## held-out labels, authored blind

eval labels were written by reading source before the ranker ever ran on them — so the eval is allowed to say the ranker is wrong. It has.

## sanitizer wall

ASan + UBSan + integer + float-cast, no-recover; 16 libFuzzer targets over its own readers of bytes it did not create; ThreadSanitizer nightly on main; leak suppression is one pinned file

## its own output is audited

the showcase capture is regenerated and adversarially re-read as a claims corpus — the methodology ships in docs/METHODOLOGY.md

// the reason to believe any number on this deck

# Claims you can trust, because we publish what failed

664

gate scripts named by test/regression.sh — and the COUNT itself is gated against the runner's own loop, so it cannot go stale quietly

8

registered NEGATIVES — changes built, gated green, measured against a band written before the code, and reverted rather than tuned

0

numbers on this deck without a committed instrument behind them — §8 lists the ones this project refuses to publish at all

## the fresh-clone arm

our own battery only ever ran inside a configured, long-lived worktree — one value of every AMBIENT input. It clones HEAD into a genuinely unconfigured checkout and re-runs the gates whose past failures were exactly that: a developer's git config, a fixture's inherited branch name, the checkout's own path length.

## the merge-conservation gate

consistency is not conservation. A conflict resolution can drop a gate from the runner's loop while regenerating the pinned count from that SAME damaged list — both sides then agree at a wrong number. On a merge, the loop must equal the UNION of both parents', minus explicit tombstones.

## the eight, by name

r7's prose-strip fix (predicted 12/22, band 10–14, measured 5/22) · RTA-lite, refuted twice · file-level evidence pooling, +0.00pp held-out · un-guarded query stemming · its IDF-guarded retry, VOIDED by its own verifier · query-term density weighting, both displacement guards tripped · --anchor and --cochange-boost, no confirmed lift · a name-coverage floor that MET its pre-registered band and was reverted anyway for a standing requirement.

*A gate that cannot go red is a gate that proves nothing — which is why every arm on this deck ships with the mutation that makes it fail.*

// own the losses — they are in the README too

# Where it loses, published with the wins

**The public C++ number is lower than the private one it replaced**

SFML: strict file@10 28.7%, any@10 41.7%, first-hit MRR 0.21. Until 2026-08-07 this bullet compared against a ~89% any@10 private corpus that is no longer reproducible. That comparison is retired; the public number is the baseline going forward.

**--grep costs more than it saves**

+19.7% tokens / -11.2% — published next to the flag it indicts, and carrying the same private-corpus caveat as every figure in that source

**PageRank is the wrong co-change ranker**

3.8% recall@5 against 40.3% for plain lexical, and fusing the two made it worse — relatedness is lexical, importance is structural

**Strict multi-file localization stays hard**

single-file gold 73.4% vs multi-file 18.2% (held-out); the same cliff on every corpus — open headroom, said plainly

*A tool that publishes its counterexamples is a tool whose wins you can believe.*

```
// the single command for "is this code actually rotten?"
```

# Six families of evidence, counted — never blended

|                   |                                                            |                             |
|-------------------|------------------------------------------------------------|-----------------------------|
| <b>structural</b> | shape: complexity, size, nesting, params, readability rank | --metrics                   |
| <b>lexical</b>    | identifier text: the naming rules                          | --lint --naming-consistency |
| <b>confusion</b>  | syntax: the atoms-of-confusion rules                       | --lint                      |
| <b>historical</b> | git change frequency, measured PER FILE                    | --hotspots                  |
| <b>colocation</b> | how much you must READ outside this file                   | --context-ratio             |
| <b>state</b>      | this function's OWN BODY touching non-local mutable state  | --nonlocal-state            |

**Ranked by the COUNT of distinct families that fire** — never a weighted composite, because a composite lets one loud family impersonate six.

**4 of 6** is what the strict preset counts — the verb says so itself, families="6" enabled\_n="4". Two families did not clear the stability ladder.

```
--quality-panel[=strict|default|lenient] — and --quality-delta for the narrower question: what did MY change make worse?
```

// a new lens does not ship because it sounds plausible

# The calibration that disqualified a family

**+0.168**

the LARGEST cross-family correlation in the whole 6×6 matrix — widening the panel from four families to six did not raise it

**27,889**

eligible functions, five independent trees — two reported separately and never pooled, because one of them is this repository

**4 of 6**

families the strict preset actually counts — TWO were measured and held back from gating

**The uncomfortable result, kept.** The stability pass ran each family across a ladder of commits to ask whether it is steady enough to gate on. **colocation came out worse than historical on one ladder**, so neither gates: strict counts four of the six — a lens its own author wanted, measured, and demoted. The ladder was run precisely because that answer was possible; assuming it would pass is how a plausible axis becomes a permanent one.

**--field-affinity was NOT made a family either** — an exclusion by unit, not a failed measurement: it ranks struct FIELDS by co-access against declared layout, and the panel's unit is a function. Stated in the calibration rather than left for someone to notice.

```
// built for the agent's seat
```

# One command wires it into your agent

```
$ ripwire wrap claude
```

claude · cursor · codex · windsurf · gemini · aider · opencode · openclaw · hermes — or --all to detect every one you have installed

## 33 MCP verbs

17 read verbs mirroring the CLI, 13 flagship reflexes (impact, uses, affected, edit\_check, from\_trace, connect ...), 3 span-addressed edit verbs with a safety contract

## lazy-body handles

read verbs return signatures and a stable handle; the agent fetches a body only when it decides it needs one — names by default, bytes on request

## 18 agent skills

moment-matched workflows (orient, navigate, change-check, quality-bar ...) — wrap prints the recipe, skills/install.sh installs them

## 12 orchestrator loops

copy-paste prompts in prompts/: run the same audit, eval and head-to-head machinery that built this tool, on your own repository

**wrap does not just print JSON:** it probes what that agent already has, emits the config in that agent's own shape, and includes a use-when blurb so the agent knows which verb fires at which moment — not just that a server exists.

the MCP server exposes the same deterministic engine — one index, shared with the CLI, staleness-checked

// in the field: one engagement, the agent's own report, an older version

# What it did for an orchestrated fleet of coding agents

## --callers

### one call redirected a task

A task was chartered to switch on a behaviour the team believed was live. One --callers call showed it had no production caller, and the task moved to the real gap: the data it needed had never been wired.

## --edit-check

### a silent divergence, caught early

A widely shared computation was refactored. --edit-check flagged incompatible call sites that a text search had missed; left alone, they would have drifted from the canonical path the day the feature was enabled.

## --quality-delta

### quality became a protocol

Run at each agent's "done", it made fix-or-justify the habit of agents that shared no context. Duplicated routines and near-identical twins were fixed, not waived.

## the honest boundary, from the same report

**The deepest findings did not come from the tool.** They came from measurement the agents built themselves: worktree diffs, attribution sweeps, funnels. The tool was a floor for orientation and honesty, not a substitute for measurement design. And one known false-positive mode: --edit-check's arity heuristic over-counts where trailing parameters have defaults, and compiling settles it, as its own notes say.

```
// it does not only read
```

# Span-addressed edits: the map writes back

## Three verbs, one resolved span

```
--replace-symbol-body=TARGET  
--insert-before-symbol=TARGET  
--insert-after-symbol=TARGET  
--edit-payload=FILE| -
```

TARGET is a name, an @FILE:LINE seed — paste the location out of a diff hunk or a compiler error and it binds the innermost enclosing definition — or a freshness-pinned handle from --grep --handles. No line numbers to keep in your head, and no whole-file rewrite.

## The refusals are the feature

Freshness hash, lock, a re-check immediately before the rename, fsync, mode preserved, atomic rename. A target that resolves to more than one definition refuses. An empty payload refuses rather than being read as a deletion. Every refusal leaves the file byte-identical, and every success prints a JSON receipt whose span is the POST-EDIT byte range — where the payload now sits, not where it was aimed.

## Preview the bytes before they exist

--edit-check=SYM --edit-payload=- --dry-run splices the payload over the span IN MEMORY, re-parses that one file, rebuilds the call graph over the re-derived tree, and answers the contract question about bytes nobody has written: did the signature change, and which callers does it break? Preview, then apply — no Read of the file in between.

## Several edits, one transaction

--edit-plan=FILE is a versioned JSON multi-edit plan and its mode is EXPLICIT: exactly one of --dry-run or --apply, and neither is not a default. Every target, payload and span is preflighted before any write; overlapping edits refuse; payload paths are confined to the plan's own directory, so an absolute path or a '..' escape refuses and names what it resolved to.

**Same engine on both seats:** the three MCP edit verbs are these three flags, and the navigation the deck has no room for is the same story — --graph-query, --whereis, --stray-content, --at, --arch, --owners, --export, --help-task all sit in docs/COMMANDS.md with a recorded invocation and its output.

every claim on this slide is stated by `ripwire --help` and captured in docs/COMMANDS.md — test/deckcheck.sh proves each flag named here exists in the shipped binary

// standing on giants

# The research inside — classic and current

49 repositories + 71 papers folded · and a labelled survey of 237 tools that contributed nothing, which says so — every row with the lesson taken and where it lives: docs/LINEAGE.md

**1976**      **Cyclomatic complexity — McCabe**  
the metric behind --hotspots

**1994**      **Okapi BM25 — Robertson · Spärck Jones**  
the lexical ranker (twice: subtoken+body, whole-name)

**1998**      **Personalized PageRank — Page · Brin**  
the headline importance signal

**2008**      **Louvain modularity — Blondel et al.**  
the module / community map

**2009**      **Reciprocal Rank Fusion — Cormack et al.**  
deterministic signal fusion (--rank-by=rrf)

**2011**      **Readability model — Posnett · Hindle · Devanbu**  
the lexical family of the quality panel

**2019**      **Delta Maintainability Model — di Biase et al.**  
--dmm: one comparable number per change

**TDAD · arXiv 2603.17973**

a static test-map cut agent-caused regressions 6.08% → 1.82%; prose TDD instructions alone made agents WORSE → the shape of --test-gate

**What-to-Retrieve · arXiv 2503.20589**

retrieval selection beats retrieval volume for coding agents → the selection-over-dumping thesis

**LocAgent / Loc-Bench (2025)**

the localization metric (strict Acc@k) and the frozen 560-instance dataset every accuracy number here is scored on

**scip-clang · Serena / clangd**

the compiler-grade oracle round 9 graded both tools against — an outside referee, not a rival

**tree-sitter**

the incremental GLR parsing substrate — 25 grammars vendored, one parser per thread

counts gated in-repo: readmedriftcheck arm E derives them from LINEAGE.md's own tables — the lineage is the design rationale

// do not take any of it on trust

# Every claim, and the command that re-derives it

185 long flags · 36 slides

```
bash test/deckclaimcheck.sh
```

every --flag named here exists

```
bash test/deckcheck.sh
```

74.7% fewer element bytes

```
bash test/showcasecapturecheck.sh
```

664 gate scripts

```
bash test/manifestcheck.sh
```

49 repos · 71 papers · 237 surveyed

```
bash test/readmedriftcheck.sh
```

the ten moments, any row

```
ripwire . --callers=SYM | wc -c
```

the head-to-head table

```
bench/headtohead/r4-2026-08-06/
```

the oracle round

```
bench/headtohead/r9-2026-08-09/RESULTS.md
```

**The deck is generated, and the generator is gated.** `present/deck5_ripwire_build.js` is scanned for fabricated flags and its slide count is derived from its own `addSlide()` calls — a deck that drifts from the binary fails the suite.

`docs/EVALS.md` is the source of truth; §7 is the counterexamples, §8 the claims this project refuses to make

```
// sixty seconds to the first ranked map
```

# Quickstart

```
git clone <repo> && cd ripwire  
cmake -S . -B build && cmake --build build -j      # hermetic: works with the network off  
./build/ripwire --help
```

```
ripwire .
```

the ranked map — start here on any unfamiliar repo

```
ripwire . --for="cache invalidation"
```

the task lens: what to touch, ranked

```
ripwire . --callers=someFunction
```

who calls it — with the honesty marker attached

```
ripwire . --quality-panel
```

is this code actually rotten? six families, one shortlist

```
ripwire . --test-gate
```

before you commit: exactly which tests must run

```
ripwire wrap claude
```

wire it into your agent, in that agent's own config shape

**ripgrep** for the speed. **tripwire** for the honesty. Apache-2.0.